

AI in VR Military Training

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ABSTRACT

Virtual Reality is the experience of which users feel immersion in a virtual simulated world through the use of hardware and software. It is then up to developers of these said worlds to create environments where the user can freely move and interact with objects or people without needing to be present physically. Training for those working in civil protection in high-pressure, dangerous and diverse scenarios in a risk-free setting is a great advantage to virtual reality training. This will help those to enhance their combat, decision-making skills, their ingenuity and overall strategic thinking. The main issues with current simulated training environments used nowadays is that the actions taken by the synthetic AI agents created are done using rule-based and purely reactive models which possess almost negligible intellect. These models are unable to adapt to external actions and surroundings and are not ideal training figures that have a sole purpose of helping users hone their skills. Another large issue is that the combinations of user actions along with the poorly made agent models still require highly intensive computational power.

Keywords

Virtual Reality, AI Agents, Reinforcement Learning, Neural Networks

1.0 INTRODUCTION

1.1 Background

To truly discuss this project, one must have a brief understanding of what VR (Virtual Reality) truly is. Virtual reality is a computer-simulated environment which could be depicted as either real or fictitious. Current VR systems are primarily visual showcased experiences displayed on a stereoscopic display. Many now also come with enhanced sensory implementations. These could include the use of more advanced haptic systems that are used to give tactile feedback. The virtual environments can be interacted with using multimodal devices such as omni directional and in this project, locations will be made to simulate scenarios present in military scenarios.

1.1 Problem Definition

Modern day virtual simulation is an import asset in military and police training but is not used to its fullest extent. There is a large room for improvement and since VR technology was not as advanced as needed in past years, the constant upgrades in performance are aiding the push towards better virtual reality training.

1.2 Background

In the past 40 years, computer generated simulated training has been incorporated greatly in the military sector, allowing military personnel the opportunity to train in a safer and more controlled setting. Unfortunately, the lack of computational strength that nowadays exists, was not present then and training was not realistic or immersive enough to replace traditional physical preparation. The implementation of virtual reality has taken this a step further by providing a more immersive and realistic experience that can truly simulate real life scenarios, giving personnel better knowledge, practice and proficiency.

1.3 Motivation for this project

The motivation I found for this project came originally from seeing the large possibilities of creations from video game engines and wanted to create something fun to work on whilst also still being incredibly useful in modern day applications/uses. The primary focus of this Final Year Project will be to experiment with different AI agents and one appealing factor to the whole project would be to eventually discuss with the possible future users of this project how it can be improved and how it could be applied further.

1.4 Aims and Objectives

The aim is to use machine learning algorithms and artificial intelligence in order to showcase improved quality implementations that would further improve modern day military and police training. The secondary aim is to create agents which users could even learn from, be it strategically, intellectually or through noticing differences in the agent's actions. This would be done by doing the following objectives. First, creating a virtual reality simulation and implementing different forms of

antagonist AI to versus the user. This is done to understand how well machine learning algorithms and artificial intelligence can be used to increase the performance and wellbeing of those involved in the military and police force. A bonus implementation that could be added is that of procedural content generation. This was discussed in more detail in the literature review.

1.5 WHY THE PROBLEM IS NON-TRIVIAL

This project has numerous challenges that are to be addressed. The first challenge being that although many AI agents have been created in both the military and video game scene, few prove to possess any substantial intellect and thus end up primarily being used as training dummies with no unique focus. The second is that strategies used in these training simulations are not easy to gain knowledge on, as one would usually have to acquire experience or communicated with well-trained individuals with strong understanding of how training is dealt with.

2.0 LITERATURE REVIEW

Below are listed the many different AI approaches that are aimed to be implemented to identify what type of agents will operate and act best under certain conditions.

Rule-Based AI

Rule-Based AI systems consist of two main aspects, these being the working memory and the rules memory. The working memory stores known facts in relation to the ruleset. The rules memory mainly consists of the if-then or if-then-else styled rules that are initiated using the known facts stored in the working memory. When rules are initiated, actions and state changes can be triggered. An example of this would be that of a finite state machine (FSM). Behaviour trees are mathematical models for plan execution which are used in computer science and artificial intelligence agents. They are used to make an object switch between the performing of specific pre-defined tasks. Their importance comes from the ability to break down complex tasks into smaller and simpler tasks without being concerned as to how implementation is done. Behaviour trees are similar to hierarchical state machines with the difference being that the key building block of a behaviour is a task instead of a state. The simplicity of these aforementioned trees makes them less likely to contain numerous errors and are used greatly in modern day video game applications.

Reinforcement Learning

Reinforcement learning is an area of machine learning which aims to take the best suitable actions to maximise rewards in a given situation or scenario. This form of learning differs from supervised learning as in supervised learning, the training data has its answer given in a key so the model can be trained to correct itself. In reinforcement learning, no answer is present so the RL agent must decide what to do for a given task based off of previous actions.

Q-Learning

This is a type of reinforcement learning policy that finds the best action given a current state. The action is chosen randomly with the aim to maximise reward. This method of reinforcement learning is model-free and off-policy algorithm that measures the quality of a state-action combination. This can thus be denoted as $Q: S \times A \rightarrow R$ in which S , A and R are the states, actions and rewards respectively. The new Q value when updating is denoted by:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha [R + \gamma \max_a Q(S_{t+1}, a) - Q(S_t, A_t)]$$

where $Q(S_t, A_t)$ = current state, α = learning rate and γ = discount factor. To learn the quality function, the algorithm uses temporal difference learning meaning that the function is updated incrementally varying on the difference between the expected reward as well as the actual reward received following an action. Gradually, the Q -function converges to the optimal action-value function. This instructs the agent which action should be taken to ensure the maximum reward is achieved. This form of reinforcement learning has been used for games, optimisations of processes and is also used in real-world situations due to its relative simplicity and power.

Neural Networks/Multi-layered Perceptron and Activation Functions

Neural networks, also known as artificial neural networks or ANNs are a subset of machine learning whose name and structure resemble that of a human brain, mirroring the way that biological neurons communicate with one another.

These networks consist of numerous layers. The first layer would be the input layer, followed by at least one hidden layer (there can be as many layers here as needed), followed by the final output layer. If a particular node/neuron, also known as an artificial neuron, outputs a value of anything above a specific given threshold value, the node is initiated and data is sent to the next layer in the network. When input data is fed into the input layer, linear transformation is conducted through the weights in the hidden layer. The result of this will then be non-linear and can then be fed into an activation function.

Activation functions are used to allow the network to learn through the inputs and outputs. The most commonly used activation functions are sigmoid, tanh and ReLU which stands for Rectified Linear Unit. ReLU is the most used out of the three has it has a relatively low computational condition as well as improved performance when solving vanishing gradient problems in comparison to the two. The main difference between the hyperbolic tangent function and the sigmoid function is that unlike the sigmoid step function, the outputs of each neuron taken for the sigmoid step function are restricted to the interval of [0, 1]. When using the hyperbolic tangent function, the output of each neuron is allowed to accept both positive and negative values, thus having an interval of [-1, 1].

$$\text{Sigmoid}(x) = \frac{1}{1+e^{-x}}$$

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\text{ReLU}(x) = \max(x, 0)$$

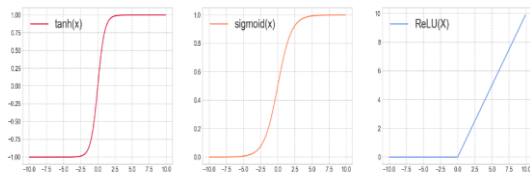


Figure 1: <https://deeptai.org/publication/reinforcement-learning-and-video-games>

From the graphs above, it can be noted that the $\tanh(x)$ function is ideal when negative values can be accepted unlike its counterparts, the $\text{sigmoid}(x)$ and $\text{ReLU}(x)$. $\text{ReLU}(x)$ only increases linearly and does not increase if a negative value is given.

Procedural Content Generation

Helps users and ai agents learn to adapt to different environments

Procedural generation is a method in which data is created algorithmically instead of manually, to combine manmade assets, terrain creation and processing power. In computer graphics, this is mainly used to create textures as well as smooth yet unique terrains. Advantages of procedural generation, varying in how its implementation is done, could be smaller sizes for files, larger amount of content, unique environments and thus, unpredictable gameplay.

It is frequently used to develop large scale virtual maps. It could save large amounts of time to negate manual map development. One key benefit to this type of map generation is that maps can possess unique features and can have near limitless permutations.

Procedural Generation is also used along with machine learning implementations to further

increase the variation of results. It is also used along with different user inputs like seed values to enable more control options for created content.

Perlin Noise

Perlin noise is a type of gradient noise developed by Ken Perlin in the 1980s. It is often used to generate procedural content in computer graphics, such as textures, terrain, and animations.

Perlin noise is a form of gradient noise that was developed by Ken Perlin in the 1980s which is mainly used for the procedural content generation for computer generated terrain, animations, images or textures. This form of noise is generated by sampling a mathematical function at frequent intervals. This function generates random values whose outputs are then interpolated to create smoothness and continuity. One key element is that Perlin noise is coherent. This means that adjacent samples would be correlated to avoid blocky and disconnected patterns.

3.0 PROPOSED SOLUTION

The proposed solution of this project is to first create virtual simulations and environments that are to be used in order to help train both users and artificially intelligent agents to improve their skills regarding specific military practices and training procedures to ensure optimal performance in the real-life scenarios that users might then have to face. This would be done by creating artificial agents using machine learning algorithms and practices in order to better their performance and realism. Since the goal is to also create AI agents whose expertise should rival or even surpass that of a human, it would also be wise to not only have users learn by practicing versus the AI, but also from the AI. To do this, machine learning approaches discussed earlier will be applied to the agents placed in specified scenarios. These agents would then be trained to complete specific tasks as effectively and efficiently as possible.

3.1 METHODOLOGY

To begin, the Unity Engine will be used to create the environments to be used for the agent's training procedures. Scripts will be made in C# to carry out the programmed tasks needed. Depending on time, different environments or even PCG generated environments could be used to diversify training and create more indifferent experiences. Training environments will otherwise be made manually so that different challenges could be presented to the user or the allied agent AI. Although the experiences will be the same each time, environments could be hand-made to ensure different strategic tasks are tackled in an ideal manner. Different external libraries for the discussed reinforcement learning and/or neural network approaches could be adopted and the varying results of the agents will be recorded to find out which work best under specified conditions. For this, libraries such as TensorFlow could prove ideal.

4 TESTING AND EVALUATION

Discuss how actual users of this kind of software can be

Mention not only can people train vs ai, but they can also learn from them – create ai agents with optimal strategies

When the training of agents is complete and all VR implementations are done, testing will be done by allowing experienced military involved users to playtest and discuss their thoughts on the final product. A series of tests can be conducted to see if performance of users would have improved as of using the software for numerous hours. Results can be gathered from both the User VS AI segments as well as the feedback from watching AI agents perform in the tasks made.

5 CONCLUSION

The progress report summarises the main machine learning implementations that will need to be implemented to pursue the research's aims and objectives as well as their respective challenges.

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Gantt Chart

	01/23	02/23	03/23	04/23	05/23
<i>Creation of Environment</i>					
<i>Implementation of VR and Game Mechanics</i>					
<i>Implementation of AI techniques for enemy agents</i>					
<i>Implementation of AI techniques for allied teacher agents</i>					
<i>Completion of Aimed Training Environment + Possible implementation of PCG</i>					
<i>Test Training Software</i>					
<i>Gather and Evaluate Results</i>					
<i>Write First Draft</i>					
<i>Write Final Draft</i>					